MULTIVARIABLE CALCULUS

Questions to Practice

- **12.4:** Questions (1–8), (15–22)
- **12.5:** Questions (1–20)
- **14.1:** Questions (1–29), (37–52)
- **14.2:** Questions (1–48)
- **14.3:** Questions (1–55)
- **14.4:** Questions (1–38, excluding 13–24)
- **14.5:** Questions (1–28)
- **14.7:** Questions (1–10)
- **15.1:** Questions (1–30)
- **15.2:** Questions (19–28)
- **15.5:** Questions (7–17)

Definitions to Know

1. Functions of Several Variables
2. Open Set
3. Closed Set
4. Interior Points
5. Exterior Points
6. Limit Points
7. Bounded Set
8. Unbounded Set
9. Continuity of a Function
10. The Mixed Derivative Theorem
11. Chain Rule for Functions of One Independent Variable and Two Intermediate Variables
12. Gradient
13. Directional Derivative
14. Local Minima
15. Local Maxima
16. Critical Point
17. Saddle Point
18. Point of Inflection in Multivariable Calculus
19. Hessian
20. Fubini’s Theorem

Python Code